SPACE COMMUNICATIONS TO AIRCRAFT: A NEW DEVELOPMENT IN INTERNATIONAL SPACE LAW (PART II)+

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The first part of this article concluded by noting the Fourth Recommendation of the International Conference on the Establishment of a Maritime Satellite System. This was that a study should be made of the possible use by INMARSAT of multi-purpose satellites providing both a maritime mobile and an aeronautical mobile communications capability.

Pursuant to this recommendation, the INMARSAT Directorate undertook technical studies of aeronautical satellite communications, and held detailed discussions with the International Civil Aviation Organization (ICAO), the International Air and Transport Association (IATA), airlines, aeronautical industry representatives and other members of the aeronautical community. The Director General reported on the results of these discussions to the Third Session of the INMARSAT Assembly, held in October 1983.[1] By this time the ICAO Council had already expressed interest in the availability of mobile satellite telecommunications for the international civil aviation community.[2] Also, INMARSAT had modified the Request for Proposals for its second generation satellites, to provide for a communications capability in the aeronautical mobile 1-1 satellite 'R' band.

The INMARSAT Assembly noted the Director General's report and requested the Director General "to study what amendments to the Convention and Operating Agreement would be desirable relative to the provision of aeronautical satellite telecommunications, and to report thereon to the next session of the Assembly."[3]

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1. ASSEMBLY/3/7.

2. ICAO doc. C-WP/7941 at 2-3.

3. ASSEMBLY/3/7/para.11.2.

The Amendment Process

The procedures established for amendment of the INMARSAT Convention and Operating Agreement are relatively lengthy.[4] Amendments may be proposed by any Party or, in the case of the Operating Agreement, by any Signatory. The proposed amendments are circulated to all Parties and Signatories for comment, after which they are considered by the Council, which in turn expresses its views to the Assembly. If adopted by the Assembly, the amendments enter into force after individual notices of acceptance have been received from a special majority of the parties.

The Director General submitted some draft amendments for initial consideration by the INMARSAT Council at its Eighteenth Session, in July 1984.[5] The Council decided to invite any willing Party and Signatory formally to initiate the amendment procedures by submitting proposed amendments to the Director General.[6] In response, amendments were submitted by both Norway and the United Kingdom. The two sets of proposed amendments differed from each other principally in their treatment of the relationship between INMARSAT and ICAO, an issue which is discussed below. Comments on various aspects of the proposed amendments were received from several other Parties and from interested bodies, including ICAO.

The proposed amendments and related comments were submitted to the INMARSAT Council for consideration at its Twentieth Session, in February and March 1985. The Council did not attempt to resolve all the outstanding issues, in particular, the relationship between the proposed amendments and the institutional competence of ICAO. Neither did the INMARSAT Council seek agreement upon a single version of the text of the amendments. However, there was general support for the objective of the proposed amendments; and the Council decided to recommend to the Assembly "the adoption of amendments to the INMARSAT Convention to enable INMARSAT to provide aeronautical satellite communications services," based on the proposed texts and the comments received.[7] The

4. INMARSAT Convention, art. 34; INMARSAT Operating Agreement, art. XVIII.

5. COUNCIL/18/5.

6. COUNCIL/18/SR/FINAL/para. 16.1.5.

7. COUNCIL/18/SR/FINAL/para.16.1.4.

relevant amendments to the Operating Agreement were approved, with a recommendation to the Assembly to confirm them.[8]

On 13 March 1985, the Director General invited all INMARSAT Parties to be represented at a Meeting of Experts, to be held at INMARSAT's Headquarters. The purpose of the meeting was to prepare a further consolidated text of the proposed amendments, for consideration and adoption by the Fourth Session of the INMARSAT Assembly. It should be noted that the convening of such a meeting is not a part of the formal amendment processs. The object was to attempt, so far as possible, to resolve disagreements relating to the amendments before the Assembly met.

The Meeting of Experts took place from 7 to 9 May 1985. Twenty parties were represented, and observers attended on behalf of IMO and ICAO. A further consolidated text was prepared for submission to the INMARSAT Assembly, together with various individual comments and statements.

The Fourth Session of the INMARSAT Assembly took place in October 1985. The remaining problems of drafting were resolved to the extent that the Assembly was able to adopt the amendments to the Convention. It also confirmed the amendments to the Operating Agreement.

The amendments to the Convention will enter into force one hundred and twenty days after the Depositary[9] has received notices of acceptance from two-thirds of those States which, at the time of adoption by the Assembly, were Parties and represented at least two-thirds of the total investment shares.[10]

The amendments to the Operating Agreement will enter into force one hundred and twenty days after the Depositary has received notices of approval by two-thirds of those Signatories which, at the time of confirmation by the Assembly, were Signatories and then held at least two-thirds of the total investment shares.[11]

It is, at present, expected that the amendments will enter into force at some time in 1988. This delay may seem surprising; but in several Member States legislative action is required before notice of approval of the amendments can be given. In certain others, the acceptance process has become linked with internal regulatory considerations. Meanwhile, INMARSAT has been actively planning and

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8. *Id.*

9. The Depositary of both the Convention and the Operating Agreement is the Secretary-General of the International Maritime Organization.

10. INMARSAT Convention, art. 34(2).

11. INMARSAT Operating Agreement, art. XVIII(2).

promoting its future aeronautical services; and has offered certain limited aeronautical services on an experimental or interim basis.

The Substance of the Amendments

As amended, article 3(1) of the INMARSAT Convention provides that the purpose of the Organization is as follows:

> to make provision for the space segment necessary for improving maritime communications and, as practicable, aeronautical communications, thereby assisting in improving communications for distress and safety of life, communications for air traffic services, the efficiency and management of ships and aircraft, maritime and aeronautical public correspondence services and radiodetermination capabilities.

The inclusion of the words "as practicable" has its origins in a concern of the United States that INMARSAT should not be obliged to provide aeronautical services, but merely permitted. The United States initially proposed that this should be made clear by an amendment to Article 3(2). The original text reads as follows:

> The Organization shall seek to serve all areas where there is need for maritime communications..

The United States proposal was to amend this as follows:

> The Organization shall seek to serve all areas where there is a need for maritime communications and may seek also to serve aircraft by providing a space segment for improving aeronautical communications...[12]

This proposal was based on an assumption that aeronautical services would also be provided by other systems competing with INMARSAT;[13] so that the Organization should not be obliged to provide services for which it had no guaranteed demand. However, the Meeting of Experts did not favor the proposed American text, preferring to deal with the matter by the use of the words "as practicable" in article 3(1). This text was adopted by the Assembly.

12. COUNCIL/20/SR/FINAL, Annex II.

13. *Id.*, Annex I.

It may be doubted whether provisions dealing with the purpose or competence of an international organization do impose on the organization or its Member States a duty to fulfill such a purpose or exercise such a competence. However, if one assumes that article 3 places a duty on INMARSAT to provide the various services referred to, it is doubtful whether the insertion of the words "as practicable" is really apt to convert such a duty into a mere competence. It would seem rather that an absolute duty becomes a qualified one. The distinction between a qualified duty and a mere competence is important. At present, for example, it could hardly be argued that it was not "practicable" for INMARSAT to offer aeronautical services. However, the point is likely to remain academic, because none of the INMARSAT Parties has opposed its plans to introduce such services.

In another respect, however, there is a much clearer distinction between the bases on which INMARSAT provides, respectively, maritime and aeronautical services. As regards maritime services, INMARSAT has a limited degree of protection from competition under article 8 of the Convention. This provides that a Party is to notify the Organization in the event that it or any person within its jurisdiction intends to make provision for, or initiate the use of, separate space segment facilities to meet any or all of the purposes of the INMARSAT space segment.[14] The stated object is "...to ensure technical compatibility and to avoid significant economic harm to the INMARSAT system." Article 8 has its origins in a concern that the market for maritime satellite communications might be modest in relation to the cost of providing the necessary space segment; and that INMARSAT might not be viable financially if unrestrained competition were allowed. The article provides for the INMARSAT Council to make a recommendation with respect to technical compatibility and for the Assembly to make a recommendation with respect to economic harm. Although both such recommendations are of a "non-binding" nature, it may be inferred that the relevant Party is at least obliged to consider such recommendations in good faith.

It was never envisaged that INMARSAT should enjoy such protection from competition with respect to aeronautical services. The international civil aviation community had made no commitment whatsoever to use INMARSAT services, and ICAO had expressly disclaimed any such commitment by itself, its Member States or users.[15] Both Norway and the United Kingdom, in their respective proposed amendments, restricted the scope of article 8 by the insertion of the word "maritime"; that is to say, the obligation to notify would only arise with

14. There are comparable provision under art. XIV of the INTELSAT Agreement.

15. ICAO doc. C-WP/8126, attachment 1, para. 2.

respect to separate space segment facilities which met any or all of the maritime purposes of the INMARSAT space segment. There was scarcely any opposition to this qualification, and in due course the Assembly adopted such an amendment to article 8.

A much more controversial issue concerns, as already indicated, relations between INMARSAT and ICAO. The relationship between INMARSAT and other international organizations is governed by article 27 of the Convention which, in its original version, reads as follows:

> The Organization shall co-operate with the United Nations and its bodies dealing with the Peaceful Uses of Outer Space and Ocean Area, its Specialized Agencies, as well as other international organizations, on matters of common interest. In particular the Organization shall take into account the relevant Resolutions and Recommendations of the Inter-Governmental Maritime Consultative Organization. The Organization shall observe the relevant provisions of the International Telecommunication Convention and regulations made thereunder, and shall in the design, development, construction and establishment of the INMARSAT space segment and in the procedures established for regulating the operation of the INMARSAT space segment and of earth stations give due consideration to the relevant Resolutions, Recommendations and procedures of the organs of the International Telecommunication Union.

Under the Chicago Convention 1944, ICAO has the competence to adopt international standards and recommended practices and procedures relating to the safety, regularity and efficiency of air navigation.[16] It is expressly provided that such standards and recommended practices and procedures may relate *inter alia* to "... communications systems and air navigation aids, including ground marking ..."[17] The question arose from the outset as to how these should fit into the scheme of Article 27 of the INMARSAT Convention. The Norwegian proposal was to amend the second sentence of article 27, as follows:

> In particular, the Organization shall take into account the relevant Resolutions, Standards and Recommendations of the

16. Convention on International Civil Aviation, art. 37. Note that one of the objectives of ICAO is to promote "safe, regular, efficient and economical air transport": *Id.*, art. 44.

17. *Id.*

> International Maritime Organization and the International Civil Aviation Organization.[18]

The United Kingdom, on the other hand, proposed that the third sentence be amended as follows:

> The Organization shall observe the relevant provisions and regulations made under the International Telecommunication Convention and the Convention on International Civil Aviation, and shall in the design, development, construction and establishment of the INMARSAT space segment and in the procedures established for regulating the operation of the INMARSAT space segment and of earth stations give due consideration to the relevant resolutions, recommendations, standards and procedures of the organs of the International Telecommunication Union and the International Civil Aviation Organization.[19]

ICAO strongly favored a wording which would require INMARSAT to observe relevant standards adopted by the ICAO Council under the Chicago Convention.[20] There was an obvious and legitimate concern that any other wording, such as that proposed by Norway, might weaken the force of ICAO standards and recommended practices.[21]

Ultimately, however, the Assembly adopted a wording very close to that originally proposed by Norway:

> In particular, the Organization shall take into account the relevant international standards, regulations, resolutions, procedures and recommendations of the International Maritime Organization and the International Civil Aviation Organization.

In other words, the position of the ICAO was assimilated to that of IMO, rather than ITU. In order to understand this, it is necessary to compare the respective constituent instruments of the three organizations in question.

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18. Annex I to Verbal Note from Director General to INMARSAT Parties (Oct. 18, 1984).

19. *Id.*

20. ICAO doc. C-WP/7941, at 4-5; INMARSAT doc. COUNCIL/20/6/ADD/1/ATTACHMENT 2.

21. *See* Milde, *ICAO*, 10 ANNALS AIR & SPACE L. 483 (1985).

The present International Telecommunication Convention[22] provides that, "Members of the Union shall have the rights and shall be subject to the obligations provided for in the Convention."[23] It further states that, "The provisions of the Convention are supplemented by the Administrative Regulations which regulate the use of telecommunications and shall be binding on all Members."[24] These obligations are not qualified. It is therefore entirely appropriate that INMARSAT should be required to "observe" the relevant provisions of the convention and regulations. However, INMARSAT is only obliged to give "due consideration" to the relevant resolutions, recommendations and procedures of ITU organs, for these do not bind ITU Member States.

In the case of IMO, its resolutions and recommendations do not bind Member States, although in practice they are highly persuasive.[25] The practice of IMO is to promote the adoption of international maritime conventions, and to recommend to States that they become Parties to such conventions. IMO has no competence to develop international standards and regulations and to impose these directly upon Member States. This is why INMARSAT is obliged only to "take into account" the relevant resolutions and recommendations of IMO.

In the case of ICAO, as has already been noted, the Organization is empowered to adopt international standards and recommended practices and procedures relating to the safety, regularity and efficiency of air navigation.[26] However, under article 38 of the Chicago Convention, individual States may depart from such international standards and procedures. A State may do so if it "deems it necessary," or if it finds it "impracticable" to comply in all respects. States therefore have a virtually unfettered right to depart from ICAO standards and procedures; although under article 38 they are required to notify the Organization of differences between their own practice and that established by an international standard. Of course, the international standards and procedures adopted by ICAO, which are designated as Annexes to the Chicago Convention,[27] have produced a high degree of uniformity in the

22. International Telecommunication Convention, Nairobi (1982).

23. Art. 2(1).

24. Art. 42(1).

25. *See* THE INTERNATIONAL MARITIME ORGANIZATION 11 (L.S. Mankabady, ed.).

26. *See supra* note 14.

27. Article 54 of the Chicago Convention provides that international standards shall "for convenience" be designated as Annexes to the Convention, but does not provide that they shall become part of the Convention.

practice of Member States in relation to air navigation. Nevertheless, numerous States have notified departures from ICAO standards pursuant to article 38. This applies not least to Annex 10 of the Chicago Convention, which contains standards relating to aeronautical telecommunications.

It was therefore considered quite inappropriate that INMARSAT should be obliged to "observe" ICAO standards and procedures; for in this event, States would assume under the INMARSAT Convention a stricter obligation to comply with ICAO standards and procedures than they had assumed under the Chicago Convention. Accordingly, the amended INMARSAT Convention obliges the Organization to "take into account" international standards and procedures of ICAO.

Nonetheless, ICAO remains the sole international body with the competence to establish international standards and procedures relating to aeronautical satellite telecommunications. It is, furthermore, very much in INMARSAT's interest that ICAO should, in due course, establish relevant standards and procedures, because INMARSAT may turn out not to be the only provider of aeronautical satellite telecommunications services. In this event, it would be highly undesirable for the various competing systems to be technically incompatible, so that airborne equipment could operate only to one system.

It may be added that the history of INMARSAT's relationship with IMO demonstrates the importance which, in the practice of INMARSAT, is attached to the obligation to "take account" of the acts of another international organization. For example, INMARSAT has been closely involved in the development of IMO's Global Maritime Distress and Safety System (GMDSS). The requirements of the GMDSS are now an important factor in planning the INMARSAT space segment.[28]

Another of the recent amendments to the INMARSAT Convention is relevant to the relationship between INMARSAT and ICAO. Article 15 originally provided that one of the powers of the INMARSAT Council is the "determination of maritime satellite telecommunications requirements." This was amended to read "maritime and aeronautical telecommunications requirements."[29]

This amendment was again opposed by ICAO, on various grounds. The initial concern seems to have been that the word "requirements" might refer to technical standards.[30] However, technical standards are dealt with in article 27 which, as has already been shown, acknowledges the competence of ICAO in this area. The only qualification is that article

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28. *See* Lundberg, *INMARSAT: Improving Maritime Communication*, MARINE POL'Y 343-5 (1984).

29. INMARSAT Convention, art. 15(a).

30. ICAO docs. C-WP/7941, at 4; C-WP/8013, Attachment 1, at 2.

15(c) empowers the INMARSAT Council to adopt criteria and procedures for approval of earth stations (including aeronautical earth stations) for access to the INMARSAT space segment. However, the exercise of this power will be subject to article 27; and there is nothing in article 15(c) to suggest that INMARSAT has any competence similar to that of ICAO to establish international technical standards.

It was also suggested from within ICAO that, if INMARSAT were allowed to determine aeronautical telecommunications requirements, the aeronautical frequency bands might become saturated with public passenger telephone traffic, to the prejudice of safety-related communications.[31] However, it is clear that INMARSAT has no competence to allocate frequency bands for particular purposes; or, in general to coordinate frequency assignments. INMARSAT merely has a responsibility, under article 28 of the Convention, to arrange for the coordination of frequencies to be used for its space segment; and for the notification of such frequencies to ITU. This may be contrasted with the position of ICAO which, although it has no competence to allocate frequency bands, has an important role in planning and coordinating the assignments of aeronautical radio frequencies, which it does in cooperation with ITU.

In addition, the concern of ICAO was expressly linked to ICAO Assembly Resolution A22-20,[32] which had recited that:

> ICAO is responsible for developing the position of international civil aviation on all matters related to the study of questions involving the use of space technology for air navigation purposes, including the determination of international civil aviation's particular requirements in respect of the application of space technology.

It is clear that the word "requirements", in this context, does not refer to technical standards. Neither, it is submitted, does it have the same meaning as in article 15(a) of the INMARSAT Convention. The "particular requirements" of international civil aviation seem to have two aspects. First, there are the particular types of communication which the industry requires, such as air traffic control and meteorological reports. Secondly, there are the essential characteristics of such services: whether a service will be voice or data; whether a high or low data rate will be used; whether the communication must be instantaneous or whether, for example, a "store and forward" message system will be adequate. INMARSAT clearly cannot determine the "requirements" of international civil aviation, understood in this sense..

31. ICAO doc. C-WP/8013 at 4.

32. ICAO doc. C-WP/8126, para. 1.

On the other hand, these "requirements" translate into demands on space segment capacity. It is necessary to determine how many aircraft will use satellite telecommunications, for what purposes, in what volume, and in what geographical regions. This leads to a calculation of the number of voice and data channels requried, which is essential for space segment planning. It is these "requirements" for aeronautical satellite telecommunications which the INMARSAT Council is empowered to determine.[33] This point may be reinforced by observing that, in the context of article 15(a), "determine" means to ascertain, not to prescribe.

Before the amendment to article 15(a) was adopted, the ICAO Council had recognized that it would be for the INMARSAT Council to determine the commercial demand for aeronautical satellite telecommunications services provided by INMARSAT, and the conditions under which such services would be provided.[34] However, this is not a sufficient description of the proper functions of the INMARSAT Council. For example, it is enough simply to know the likely volume of air traffic control communications. The geographical characteristics of established air routes and air traffic control regions may influence the location of satellites or the type of "footprint" required. To take another example: different services may have differing requirements as to availability and reliability. This may affect whether the service is provided on a dedicated satellite or through a transponder on a hybrid satellite. It will certainly affect the question of whether and how a spare (backup) satellite is provided. These are obviously matters for the space segment provider to determine. It is difficult to think how better to describe all relevant matters than by the use of the word "requirements," although the potential for ambiguity is to be regretted.

It is entirely understandable that ICAO should insist on preserving its various competences under the Chicago Convention. Equally, INMARSAT must remain responsible for planning and providing its own space segment. This is particularly important because ICAO has expressly disclaimed any commitment to the future operational use of a particular satellite telecommunications system; and has to date refused to accept any financial commitment in relation to such a system.[35] INMARSAT has an obligation to operate on a sound financial basis, having regard to accepted commercial principles.[36] A determination of aeronautical satellite telecommunications requirements, within the

33. *See* also Lundberg, *INMARSAT - A bird in the hand?*, 41 ICAO BULL, 12 (Dec. 1986).

34. ICAO doc. C-WP/8126, para. 4.

35. See *supra* note 15.

36. INMARSAT Convention, art. 5(3).

meaning of article 15(a), is essential for the Organization to determine the feasibility and necessary features of its aeronautical satellite telecommunications system.

It can therefore be seen that, in adopting the various amendments to the Convention, the INMARSAT Assembly was not motivated by any desire to trespass upon the competence of ICAO. It is the writer's view, for the reasons set out above, that the amendments do not, correctly interpreted, have such an adverse effect. Moreover, when adopting the amendments, the Assembly also expressly recognized the competence of ICAO in the field of aeronautical communications, and resolved to recommend to the Council that it should authorize the Director General to negotiate an agreement between INMARSAT and ICAO. In particular:

> The agreement will provide for the establishment of means to ensure proper coordination and cooperation between the two organizations in all matters related to the planning and provision of the space segment used for aeronautical telecommunications.[37]

The two organizations have now reached an advanced stage in the negotiation of such an agreement.

Further Developments

INMARSAT has been an active contributor to the work of ICAO's Special Committee on Future Air Navigation Systems (FANS). This was established in November 1983 by the ICAO Council to report, *inter alia,* on technical, operational, institutional and economic questions relating to future potential air navigation systems and the possible benefits of satellite technology for international civil aviation.[38] FANS has already considered in some detail the technical characteristics of an aeronautical satellite system; but of particular interest in the present context are the initial conclusions of FANS with respect to institutional aspects.[39]

FANS has reiterated that ICAO is the only appropriate body to establish technical standards for international aeronautical communications and surveillance services. It also recognizes that States will continue to authorize communications and surveillance services in the airspace for which they are responsible. Services should be accessible to

37. ASSEMBLY/4/10/para. 7.10.

38. Report of the Second Meeting of the Special Committee on Future Air Navigation Systems (April 1985).

39. Report of the Third Meeting of the Special Committee on Future Air Navigation Systems, 6 (Nov. 1986).

all users without discrimination. Service providers must at least comply with ICAO minimum standards of performance for data and voice communications used to provide safety services. There should be an equitable attribution and distribution of costs of joint arrangements among participating States and users. Arrangements must provide for the necessary levels of safety and continuity; for reasonable pricing of services; and for State and user participation in the planning of services.[40]

FANS took into account the possible benefits of competition in the provision of services but considered also that:

> to the extent possible existing government or inter-governmental organizations should be utilized to perform the functions which may be needed. Aeronautical interests must be fully protected in such arrangements.[41]

The FANS Committee has also considered the implications for aeronautical satellite telecommunications of Article 30 of the Chicago Convention, which deals with aircraft radio equipment. Paragraph (a) of this Article provides *inter alia* that, "The use of radio transmitting apparatus in the territory of the contracting State whose territory is flown over shall be in accordance with the regulations prescribed by that State." The FANS Commitee noted that:

> individual Contracting States have enacted national regulations which may effectively prohibit any radio transmissions from aircraft operating in or over national territories, for purposes other than ATS, and related air navigation applications or on radio frequencies other than those specifically notified ... for aeronautical mobile safety services....[42]

Paragraph (b) of article 30 provides that radio transmitting apparatus may be used only by licensed members of the flight crew. The FANS Committee noted that it would appear to prevent passengers from using aeronautical public correspondence services.[43] Accordingly, the FANS Committee made the following recommendation:

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40. *Id.*, sec. 6.2.7.

41. *Id.*, sec. 6.2.5, 6.2.6.

42. *Id.*, sec.6.2.9.2.

43. *Id.*, sec. 6.2.9.3.

> That ICAO, in the light of the planned implementation of aeronautical administrative communications and aeronautical passenger communications services, as a matter of urgency, study the implications of Article 30(a) and (b) of the Convention on International Civil Aviation, with a view to enabling on a global basis, non-safety air-ground communications.[44]

It remains to note one further development affecting the international legal framework of aeronautical satellite telecommunications. In September 1987, the World Administrative Radio Conference (WARC) for the Mobile Service meets in Geneva to consider *inter alia* the frequency requirements of mobile-satellite services. Among the matters to be considered are proposals that part of the aeronautical 'R' band be reallocated or shared with other mobile-satellite services. The WARC will also decide whether to make any allocation for aeronautical public correspondence services, which are at present excluded from the aeronautical 'R' band.[45] The claims of aeronautical satellite services will be weighed against other demands on the same spectrum: for example, for land mobile-satellite services. The decisions of the WARC will be crucial to the development of satellite telecommunications services for the international civil aviation community.

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44. *Id.*, sec. 6.2.9.4.

45. *See* Lundberg, *Between a rock and a hard place,* TELECOMM. POL'Y 3 (March 1987).